Prateek Prakash Bolagond

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Professional Summary

Electronics and Communication Engineering student with a strong interest in Digital IC Design and RTL development. Proficient in Verilog HDL for designing and verifying digital systems, including FIFOs, communication protocols, and finite state machines, with hands-on exposure to synthesizable RTL suitable for FPGA and ASIC flows. Solid understanding of digital logic design, timing fundamentals, and CMOS device behavior. Seeking a Digital Design/RTL Engineering Internship to apply my skills and gain practical industry experience.

Education

University of Visvesvaraya College of Engineering (UVCE) 2023 – 2027
Bachelor of Technology in Electronics and Communication Engineering, CGPA: 8.18/10 *Bengaluru, Karnataka*

Tungal Science Composite P U College 2021 – 2023
Pre-University Course (Class XII), 95.0% *Jamkhandi, Karnataka*

DPK Central School (CBSE) 2018 – 2021
10th Grade, 89.02% *Bagewadi, Karnataka*

Technical Skills

Programming: Verilog HDL, C, Java (Basics)
Analog Fundamentals: Circuit Theory, Electronic Devices & Circuits, Analog Electronics, CMOS Fundamentals
Digital Fundamentals: Combinational & Sequential Logic, Flip-Flops, Counters, FSM, STA (basics), RTL Design
EDA Tools: LTspice XVII, ModelSim, MATLAB (basics)

Projects

Asynchronous FIFO Design | *Verilog RTL* 2026

- Designed an asynchronous FIFO with independent write and read clock domains.
- Implemented binary-to-Gray code conversion for read and write pointers.
- Used two-flip-flop synchronizers to transfer Gray-coded pointers across clock domains.
- Developed full and empty flag logic and verified FIFO behavior using a Verilog testbench.

UART Protocol Design | *Verilog HDL* 2026

- Designed a modular UART Controller featuring a configurable baud rate generator and separate transmitter and receiver modules for reliable serial communication.
- Implemented finite state machines (FSMs) for serial data transmission and reception and verified functionality through RTL simulation.
- Developed synthesizable Verilog RTL suitable for FPGA and ASIC design flows.

5T Operational Transconductance Amplifier (OTA) Design | *Cadence Virtuoso, Spectre, 90nm* 2026

- Designed a 5-transistor CMOS OTA in 90nm technology, implementing a differential input stage with a current mirror active load.
- Performed comprehensive DC, AC, Transient, and Operating Point analyses using Spectre to evaluate gain, phase margin, slew rate, and output swing.
- Optimized transistor device dimensions and sizing ratios to maximize key analog performance parameters while successfully reducing overall power consumption.

Certifications

- **Automateum 2.0** – Hands-on IoT Workshop, BMSCE IEEE PES & Sensors Council, November 2024
- **Basics of VLSI Design** – Campus Champs, 2025